

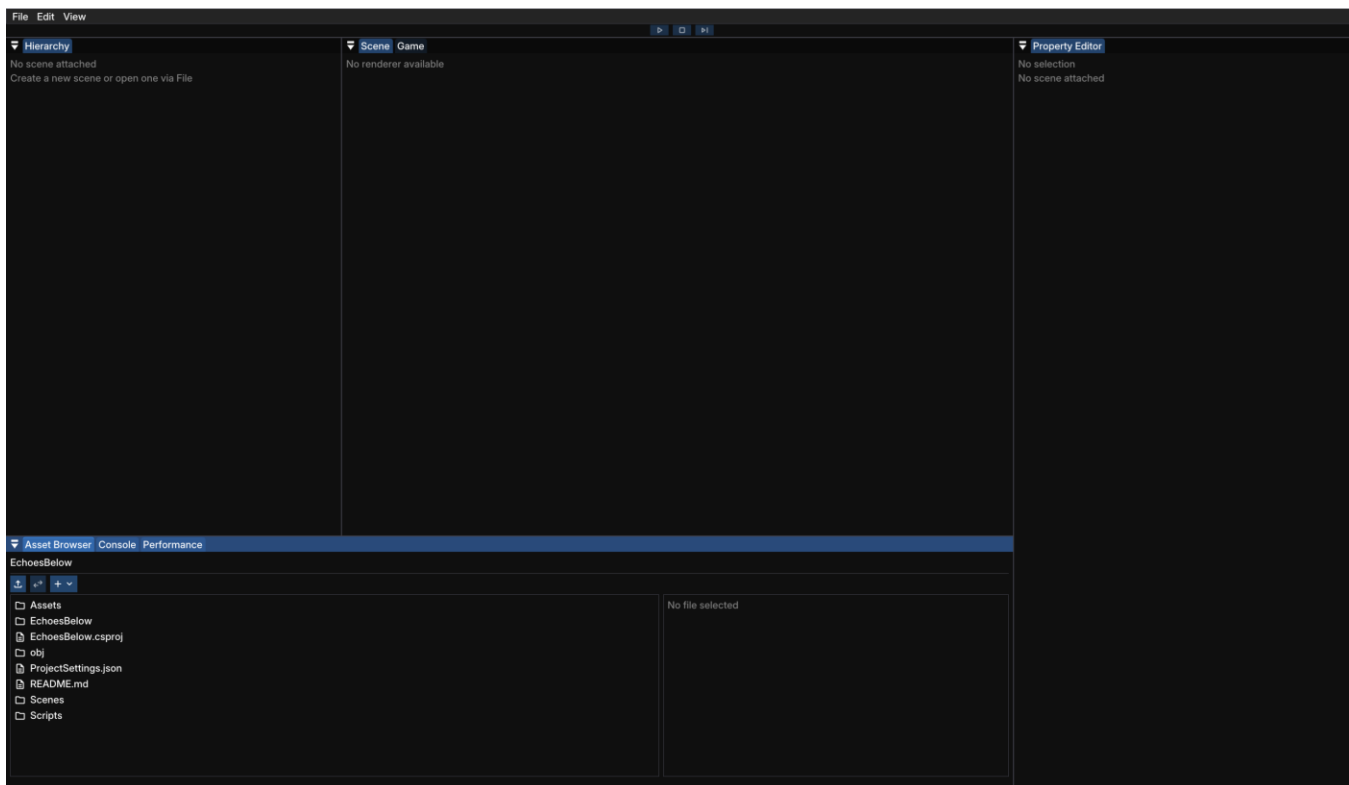
GrapeEngine Level Editor Guide

Explains the usage of the editor panels (Hierarchy, Inspector, Asset Browser, Scene/Viewport, Playback Controls, Prefab Editor, Console) and common workflows like adding entities and assigning assets.

Editor Layout & Docking

The editor uses ImGui docking to organize panels:

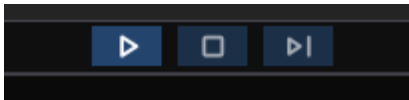
- Top toolbar: Game Controls (Play/Pause/Stop, Step)
- Left: Hierarchy
- Center: Scene (viewport) and Game (scene camera output)
- Right: Property Editor and Prefab Editor
- Bottom: Asset Browser, Console, Performance



Playback Controls (Play/Pause/Stop)

- ♦ **Play:** Saves a snapshot and starts simulation
- ♦ **Pause:** Freezes time (timeScale = 0)
- ♦ **Stop:** Restores snapshot; returns to edit state
- ♦ **Step:** Advances one frame while paused

Hotkeys: **Ctrl+P** (Play/Stop), **Ctrl+Shift+P** (Pause/Resume), **Alt+P** (Step)

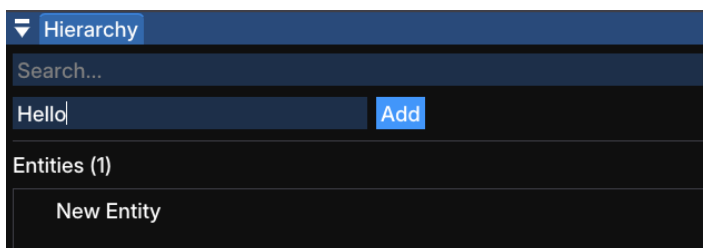


Hierarchy Panel

Lists all entities with parent/child relationships. Supports selection, search/filter, context menu actions, reparenting (drag-drop), prefab instantiation, and creation.

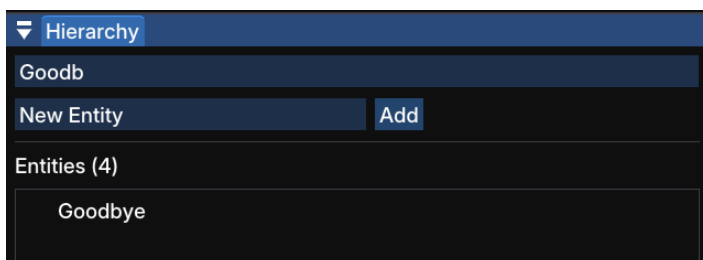
Create a New Entity

1. Open **Hierarchy** panel.
2. Enter a name in the input field (defaults to "New Entity").
3. Click **Add** to create a root-level entity.
4. Drag the entity to reparent under another node.



Search & Count

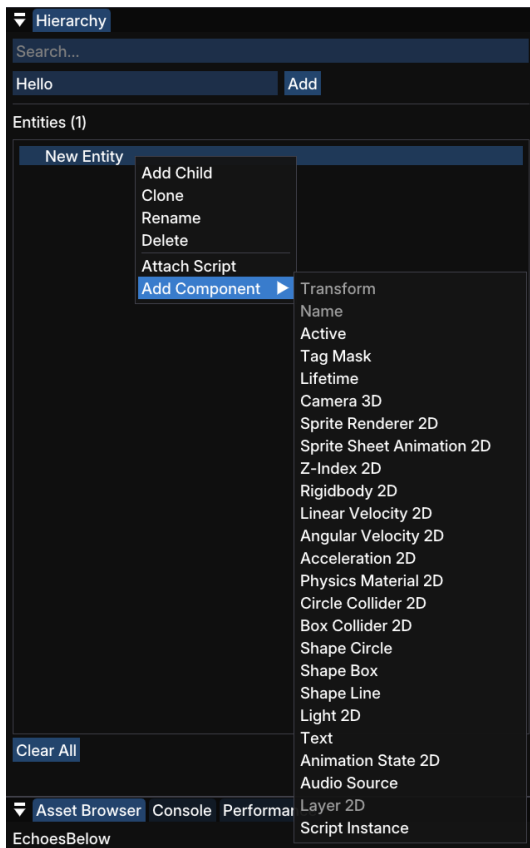
- Use the search field to filter by name.
- Entity count excludes hidden/system entities.



Context Menu Actions

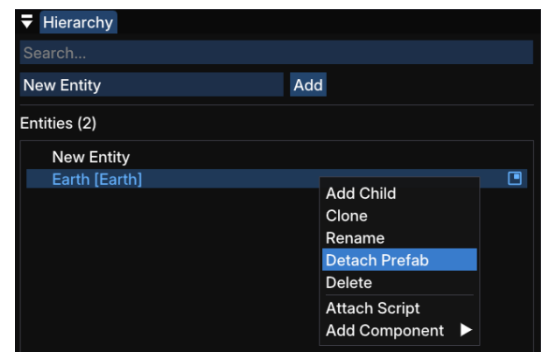
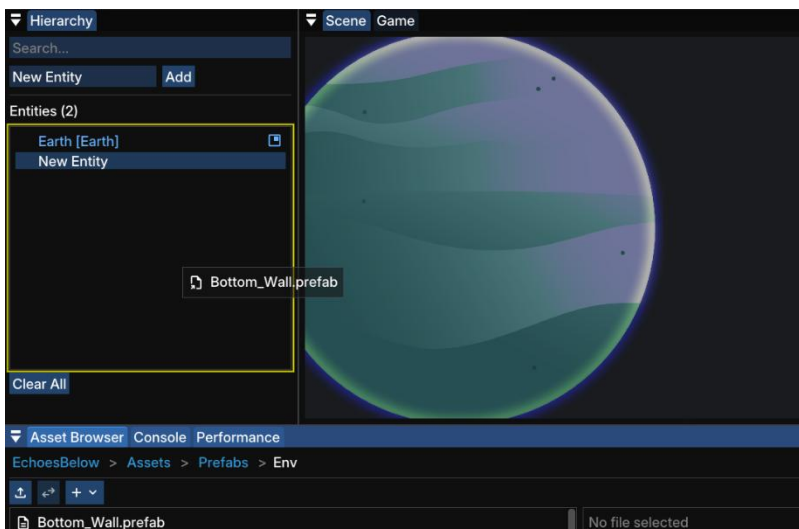
- **Add Child** (creates a new child entity under the selected parent)
- **Clone** (duplicates each selected entity, including its components and hierarchy)
- **Rename** (changes the name of the selected entity)
- **Detach Prefab** (breaks the link to the source prefab)

- **Delete** (removes the selected entity from the hierarchy list)
- **Attach Script / Detach Script** (Windows file dialog to import or remove scripts)
- **Add Component** submenu with available component types



Prefab Instantiation

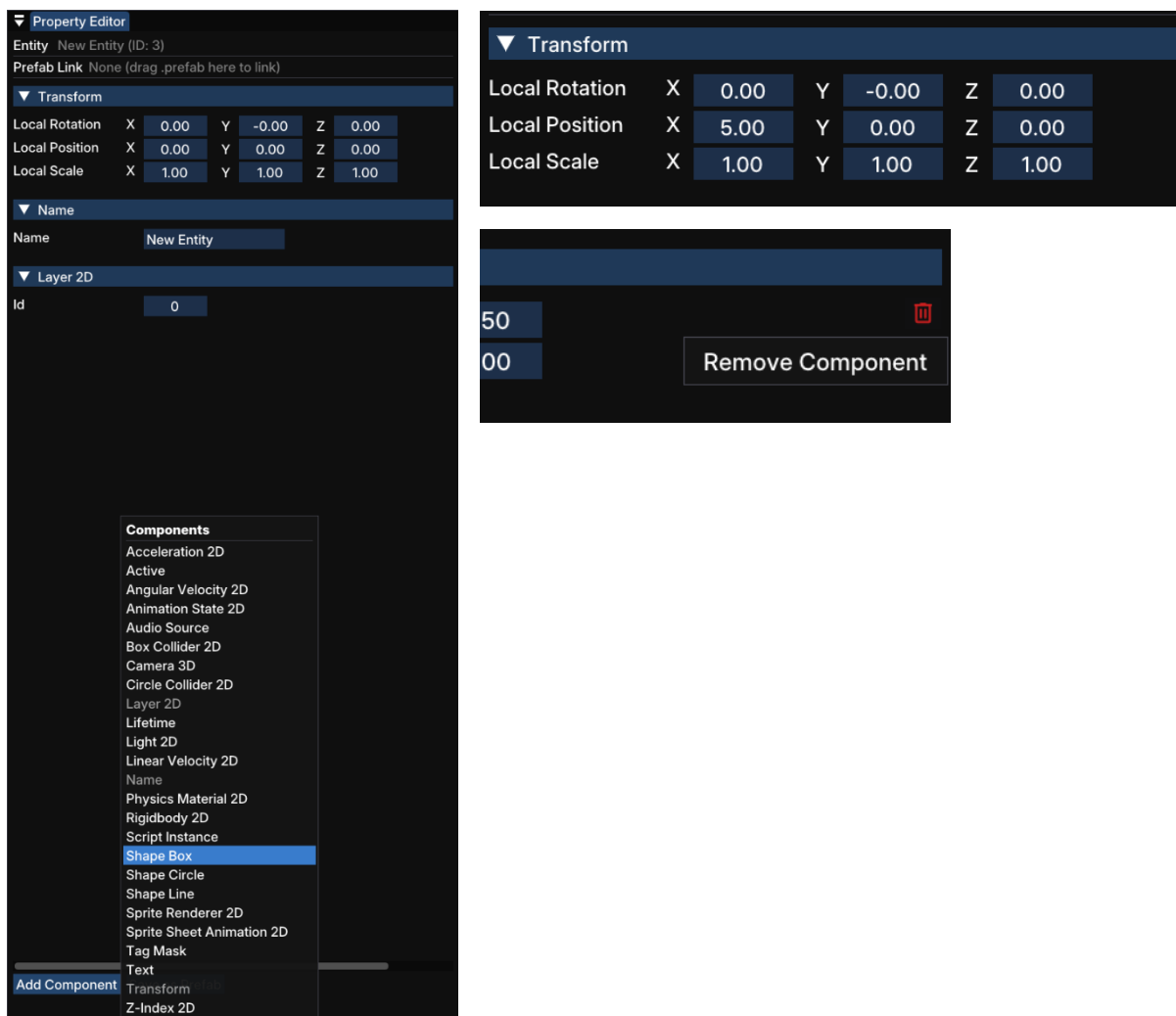
- Drag a `.prefab` onto a node to instantiate as a child.
- Instances are linked with a `PrefabLink` component and can be detached.



Inspector (Property Editor)

Select an entity to edit its components. Use **Add Component** to add new ones. Edits apply immediately and are reflected in the scene; changes are applied back to components after the UI pass.

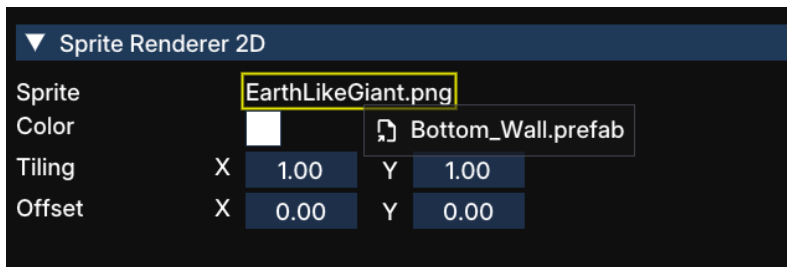
- Edit **LocalTransform** (Position, Rotation, Scale)
- Edit rendering, physics, and custom script components
- Undo records transform changes automatically
- **Add Component** popup sorted alphabetically
- Safe deletion via deferred remove (applied after UI loop)



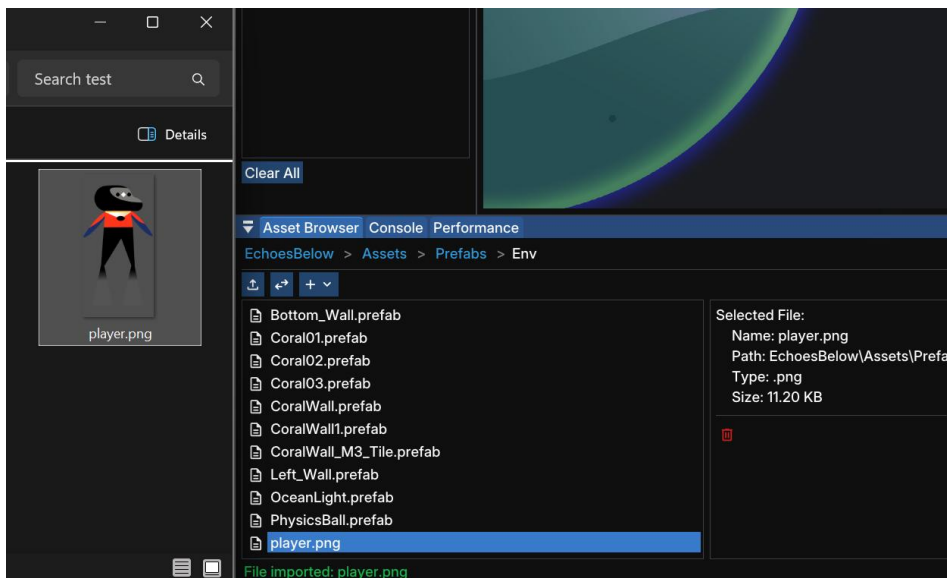
Asset Browser

Browse assets, import new files, and drag-drop into components.

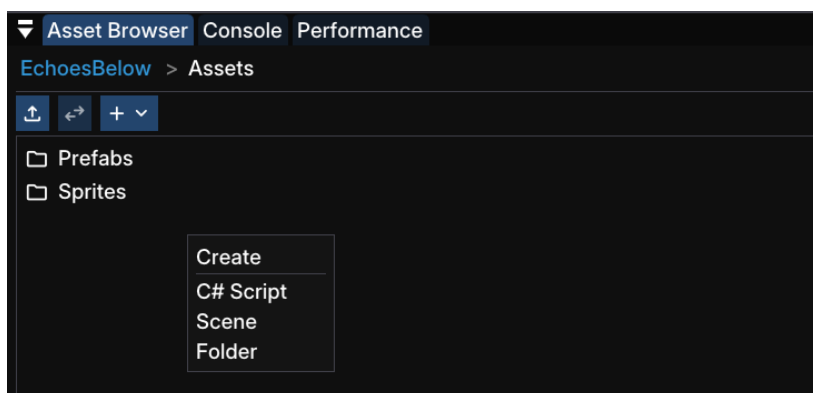
- **Internal drag-drop:** Drag a texture onto a SpriteRenderer field or audio onto AudioSource



- **External drop:** Drag files from Windows Explorer into the editor window to import



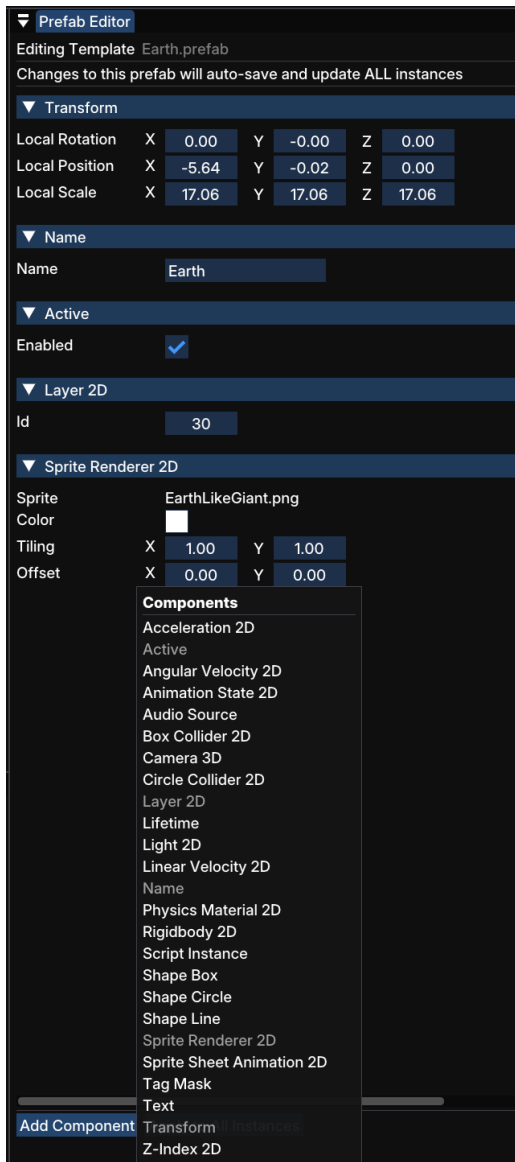
- Create assets: Right-click or use the **Create** menu (Script, Scene, Folder)
- Breadcrumb navigation at the top; multi-select with status feedback Context menu:
 - Rename (F2), Copy, Cut, Paste, Delete
 - Folder drop target: copy/move dragged assets; hold **Ctrl**
 - Prefabs: **Load Prefab** into scene, **Edit Prefab** in editor to copy



Prefab Editor

Edit prefab components and save; apply changes to new and existing instances.

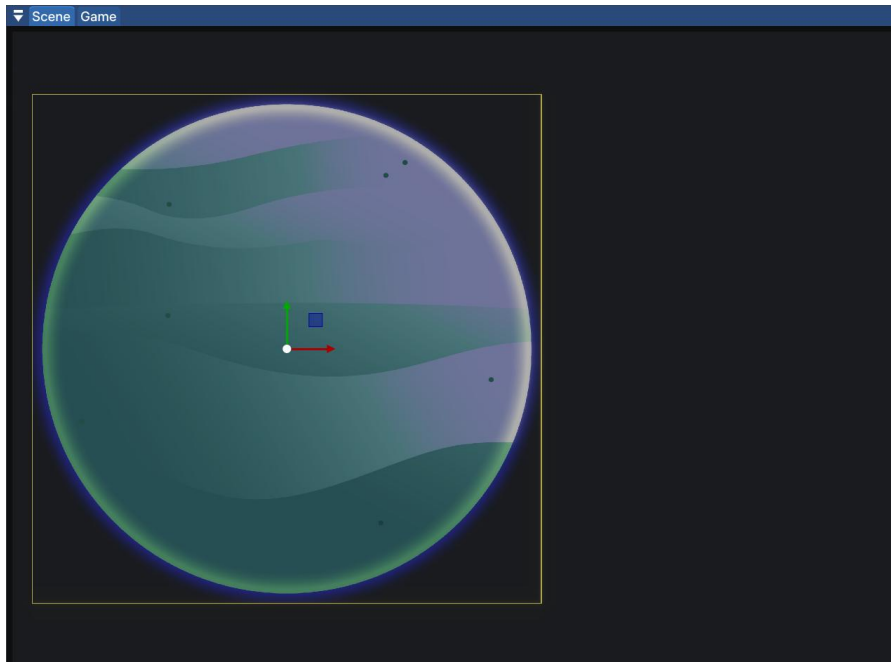
- ♦ Open a `.prefab` from Asset Browser and choose **Edit Prefab**
- ♦ Use **Add Component** and edit properties; changes auto-save
- ♦ Apply changes to all instances via prefab actions
- ♦ Components are displayed with Transform first, Name second, then alphabetical



Scene / Viewport

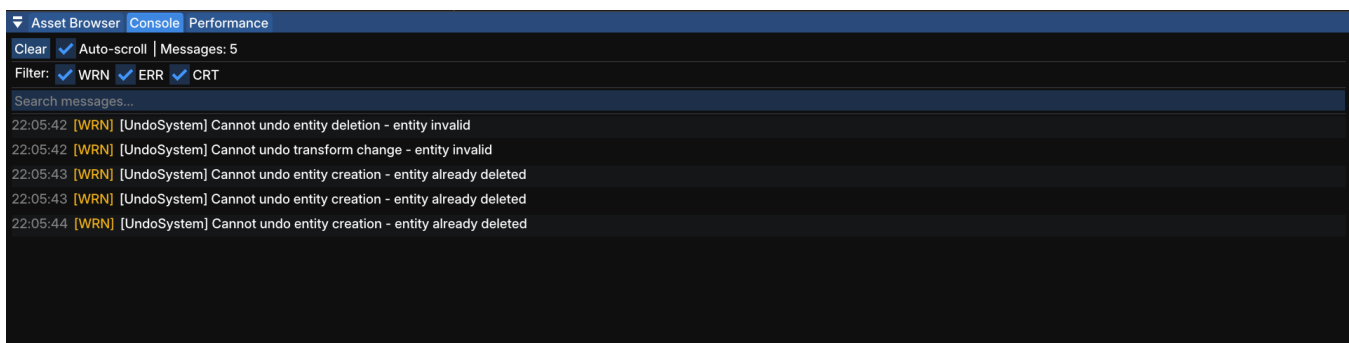
The **Scene** viewport shows the editor camera view; the **Game** tab shows the scene camera output.

- ◆ Resize the viewport; camera aspect updates automatically
- ◆ Selection in viewport syncs with Hierarchy
- ◆ Overlay shows FPS and flush count
- ◆ Tone-mapped LDR preview ensures colors match game output



Console & Performance

Use **Console** for logs and **Performance** for frame statistics.



Common Workflow: Assign a Texture

1. Select an entity with `SpriteRenderer2D` .
2. In `Asset Browser` , select a `.png` file.
3. Drag the file onto the `Sprite` field in `Property Editor` .
4. Texture path and metadata are set; scene updates immediately.

Tips

- ♦ Use Play/Pause/Stop for real-time testing; Stop restores your edit state.
- ♦ Scenes/prefabs are JSON; assets can be added without recompiling.